

Exercise 1

Consider the scalar linear discrete-time system

$$x(k+1) = (2+a)x(k) + u(k) \quad (1)$$

with unknown parameter $a \in \mathbb{R}$, $|a| \leq a_u$. The system is constrained as follows:

$$x(k) \in \mathcal{X}, \quad u(k) \in \mathcal{U}, \quad \mathcal{X} := [-5, 5], \quad \mathcal{U} := \mathbb{R}.$$

1. Provide the disturbance set $\mathcal{W}$ such that for $w(k) \in \mathcal{W}$

$$x(k+1) = 2x(k) + u(k) + w(k) \quad (2)$$

captures the uncertainty due to not knowing a .

In the following, consider system (2) with $|w(k)| \leq 2$.

2. Calculate a linear state feedback gain $K_{\mathcal{E}}$, such that the minimal robust invariant set for system (2) under $u(k) = K_{\mathcal{E}}x(k)$ is given by $\mathcal{E} = [-3, 3]$.
3. Assume $K_{\mathcal{E}} = -2$. Compute the minimal robust invariant set $\mathcal{E}$ for system (2) under the control law $u(k) = K_{\mathcal{E}}x(k)$ as well as $\bar{\mathcal{X}} = \mathcal{X} \ominus \mathcal{E}$ and $\bar{\mathcal{U}} = \mathcal{U} \ominus K_{\mathcal{E}}\mathcal{E}$.

Hint 1: $[a, b] \oplus [c, d] = [a+c, b+d]$, $[a, b] \ominus [c, d] = [a-c, b-d]$

Hint 2: $\sum_{i=0}^{\infty} q^i = \begin{cases} \frac{1}{1-q}, & |q| < 1, \\ \infty, & \text{otherwise.} \end{cases}$

Solution:

- i) We know that the system state $x(k)$ is constrained to $\mathcal{X} = [-5, 5]$. To design the additive disturbance set $\mathcal{W}$ we consider the worst case of the term $ax(k)$ w.r.t. to the state constraints, i.e.,

$$-a_u 5 \leq ax(k) \leq a_u 5.$$

Choosing the additive disturbance $w(k)$ with $\mathcal{W} = [-5a_u, 5a_u]$ ensures to capture the effect of the unknown parameter a .

- ii) Minimum robust positive invariant (mRPI)

$$E = \bigoplus_{i=0}^{\infty} (A + BK_{\mathcal{E}})^i \mathcal{W}$$

Here, using Hint 2, we have

$$\begin{aligned} [-3, 3] &= \sum_{i=0}^{\infty} (2 + K_{\mathcal{E}})^i [-2, 2] \\ &= \begin{cases} \left[-\frac{2}{1-2-K_{\mathcal{E}}}, \frac{2}{1-2-K_{\mathcal{E}}} \right], & |2 + K_{\mathcal{E}}| < 1 \\ [-\infty, \infty], & \text{else} \end{cases} \end{aligned}$$

Hence, for $|2 + K_{\mathcal{E}}| < 1$, we obtain

$$3 = \frac{2}{1-2-K_{\mathcal{E}}} \quad \Leftrightarrow \quad \frac{2}{3} = 1-2-K_{\mathcal{E}} \quad \Leftrightarrow \quad K_{\mathcal{E}} = -\frac{5}{3}.$$

Second condition leads to the same value, i.e., it follows that $K_{\mathcal{E}} = -5/3$.

iii) Obtain closed-loop dynamics under $K_{\mathcal{E}} = -2$ as

$$x(k+1) = 2x(k) - 2x(k) + w(k) = w(k). \quad (3)$$

Since there are no input constraints, it holds that $x(k+1) \in \mathcal{W}$ for all $x(k) \in \mathcal{X}$. Thus, the minimum RPI set under $K_{\mathcal{E}} = -2$ is given by $x(k) \in \mathcal{W}$, i.e., $\mathcal{E} = [-2, 2]$. Using Hint 1, the tightened constraint sets are computed as

$$\begin{aligned} \bar{\mathcal{X}} &= \mathcal{X} \ominus \mathcal{E} = [-5, 5] \ominus [-2, 2] = [-3, 3]. \\ \bar{\mathcal{U}} &= \mathbb{R} \ominus \mathcal{E} = \mathbb{R}. \end{aligned}$$

Exercise 2

Consider the linear system

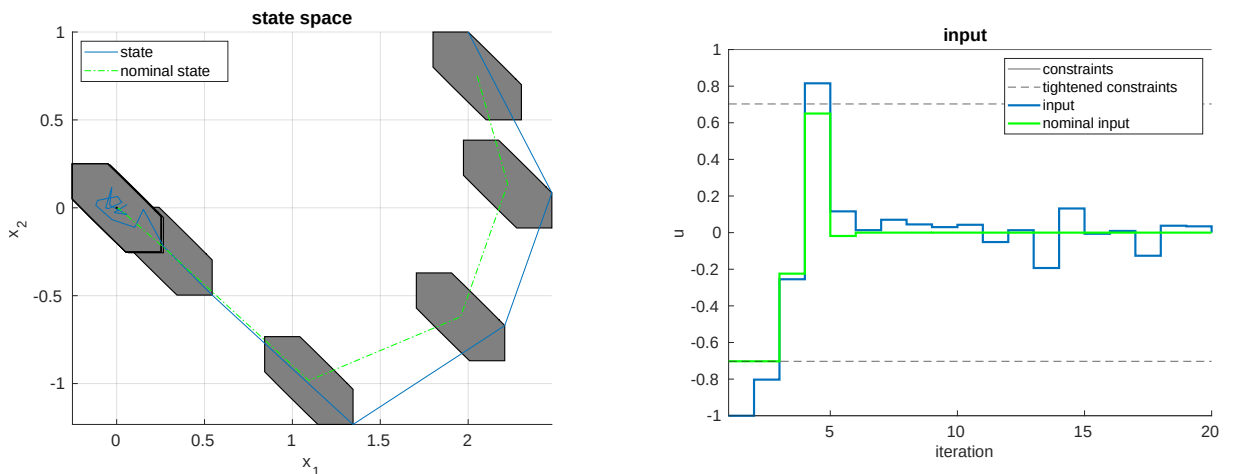
$$x(k+1) = \underbrace{\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}}_A x(k) + \underbrace{\begin{pmatrix} 0.5 \\ 1 \end{pmatrix}}_B u(k) + w(k)$$

with constraints $x(k) \in \mathcal{X} := \{x \in \mathbb{R}^2 : [0 \ 1]x \leq 2\}$, $u(k) \in \mathcal{U} := \{u \in \mathbb{R} : |u| \leq 1\}$, and $w(k) \in \mathcal{W} := \{w \in \mathbb{R}^2 : \|w\|_{\infty} \leq 0.1\}$. Let $\ell(z, v) = (1/2)z^T Q z + (1/2)v^T R v$ with $Q = I$, $R = 0.01$. Design and implement a tube-based MPC controller for the system as outlined below.

- i) Determine a valid $K_{\mathcal{E}}$ and the corresponding minimal robust positively invariant set $\mathcal{E}$.
- ii) Determine a valid terminal controller K_f and a corresponding terminal set $\mathcal{X}_f$.
- iii) Implement an MPC algorithm for the problem above with prediction horizon $N = 5$, in particular think of the following aspects:
 - Which are the variables to optimize over?
 - How can the constraints be implemented?
 - What can you observe about the initial state z_0 at each iteration?
- iv) Approximately compute the feasible set of the MPC and plot it for different planning horizons

Solution:

See MATLAB script for the design and implementation of the tube-based MPC controller. The plots below show the simulation results for the system under the tube-based MPC controller with initial condition $x(0) = [2 \ 1]^T$.



Exercise 3

Consider the nominal linear system

$$x(k+1) = Ax(k) + Bu(k), \quad (4)$$

with (A, B) controllable, and the uncertain linear system

$$x(k+1) = Ax(k) + Bu(k) + w(k), \quad (5)$$

with the same A, B matrices and unknown but bounded disturbances $w \in \mathcal{W} \subset \mathbb{R}^n$ with $0 \in \mathcal{W}$. Both systems (4) and (5) are subject to state constraints $x(k) \in \mathcal{X}$ and input constraints $u(k) \in \mathcal{U}$, for all $k \geq 0$.

Consider the standard nominal MPC controller κ defined by the optimization problem $P_{S,1}$ as introduced in the supplementary material S.1, satisfying Assumptions S.1-1, S.1-2, and S.1-3, for system (4), and the standard tube-based MPC controller κ_{tube} defined by the optimization problem $P_{S,2}$ as introduced in the supplementary material S.2, satisfying Assumptions S.2-1, S.2-2, and S.2-3, for system (5).

	True	False
The robust tube-based MPC problem $P_{S,2}$ is recursively feasible for all disturbances $w(k) \in \mathcal{W}$.	☐	☐
Suppose $P_{S,2}$ is feasible for $x(k)$ with optimal input sequence $V^*(x(k))$. Is $V^*(x(k))$ guaranteed to be a feasible input sequence for $P_{S,1}$?	☐	☐
Consider the closed-loop system (5) under $u(k) = \kappa_{\text{tube}}(x(k))$. It holds that $\sum_{i=0}^{\infty} \ell(x(i), u(i)) < \infty$.	☐	☐
If $\mathcal{U}$ is a polytope and $\mathcal{E}$ is a polytope, then $\mathcal{U} \ominus K_{\mathcal{E}}\mathcal{E}$ is a polytope if it is a non-empty set.	☐	☐
Consider a minimal robust invariant set $\mathcal{E} \subseteq \mathcal{X}$ for system (5) under the controller $u(k) = K_{\mathcal{E}}x(k)$ with $K_{\mathcal{E}}\mathcal{E} \subseteq \mathcal{U}$, and replace the terminal constraint in $P_{S,1}$ with $x_N \in \mathcal{E}$. The MPC problem $P_{S,1}$ is recursively feasible for the closed-loop system (4) with $u(k) = \kappa(x(k))$.	☐	☐
Consider a minimal robust invariant set $\mathcal{E} \subseteq \mathcal{X}$ for system (5) under the controller $u(k) = K_{\mathcal{E}}x(k)$ with $K_{\mathcal{E}}\mathcal{E} \subseteq \mathcal{U}$, and replace the terminal constraint in $P_{S,1}$ with $x_N \in \mathcal{E}$. The origin is guaranteed to be an asymptotically stable equilibrium for the closed-loop system (4) with $u(k) = \kappa(x(k))$.	☐	☐

Solution:

- True. We can construct a feasible candidate solution.
- False: Since the initial nominal state in P_2 is not necessarily equal to $x(k)$, it is not guaranteed that the terminal constraint is satisfied in the nominal problem.
- False: We only converge to a neighborhood of the origin and, e.g., a constant disturbance can lead to infinite cost.
- True: By definition of Pontryagin difference.
- True: Nominal candidate input sequence $\{u_1^*, \dots, u_{N-1}^*, K_{\mathcal{E}}x_N^*\}$ leads to recursive feasibility.
- False: Because cost decrease with respect to terminal cost is not related to terminal set $\mathcal{E}$.